

Luis Alejandro López Reyes

IT Engineering Student (3rd Year) | Full-Stack Developer

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Professional Summary

Advanced Information Technology Engineering student (UTN) and IT Associate Degree graduate with a specialized focus in Process Automation, Network Infrastructure (Cisco CCNA), and Cybersecurity. Hands-on experience architecting robust and decoupled enterprise solutions, implementing automation via relational/NoSQL databases, role-based access control (RBAC), and securing containerized environments. Proficient in developing optimization scripts and system hardening within Linux CLI environments.

Technical Certifications & Infrastructure

Cisco Networking Academy: Cybersecurity & CCNAv7: Introduction to Cybersecurity (2026), Complete CCNAv7 (Modules 1-3) (2024-2025), IT Essentials (2021).

Telecommunications Infrastructure (120 Hours): Fiber Optic and Structured Cabling Certifications - INA (2022).

Technical Skills

Automation & Backend: Node.js, Express, Python Scripting, TypeORM, REST APIs, GraphQL.

Cybersecurity & Infrastructure: Perimeter Rules, RBAC, Linux CLI (Ubuntu, Kali Linux), Network Security, TCP/IP.

Databases & Auditing: PostgreSQL (Triggers, Views, Stored Procedures), MySQL, MongoDB, SQL Server.

DevOps & Cloud: Docker, Docker Compose, Firebase (Cloud Functions, Auth, Firestore), AWS, Azure, Git/GitHub.

Programming Languages: TypeScript, JavaScript, Java, Python, SQL, Kotlin.

Frontend & Mobile Development: React 19, Vite, Tailwind CSS, React Native, Flutter, Native Android.

Featured Engineering Projects

Web-Based Point of Sale (POS) System - Multiservicios Ases y Reyes (Private Repository) – Full-Stack Developer

- Architected and deployed a web system under a decoupled modular structure, automating sales workflows, inventory logistics, and employee management.
- Designed and implemented triggers, stored procedures, and native views directly in PostgreSQL to automate internal auditing and user action logging.
- Developed a modular backend in Node.js/Express with TypeScript using the Controller-Service pattern, ensuring data persistence and isolation of sensitive enterprise data.
- Integrated a secure reactive frontend implementing encrypted JWT authentication and automated bulk data ingestion via Excel.

Kalova - Serverless Web Platform & Digital Ecosystem | kalova-898bf.web.app – Software Engineer

- Designed and deployed a serverless e-commerce platform integrated with strict data security rules and backend automation.
- Implemented a backoffice control panel secured through Role-Based Access Control (RBAC) and advanced perimeter rules in Firestore to mitigate unauthorized access.
- Programmed backend automations via Firebase Cloud Functions for secure image processing and execution of automated transactional emails.
- Developed a modular multi-step quote wizard featuring strict client-side and server-side data type validations.

Texas Trailers CR - Digital Ecosystem | texastrailerscrweb.web.app – Full-Stack & Mobile Developer

- Developed a hybrid digital solution (B2C/B2B) oriented towards logistics automation to synchronize physical inventory in real-time between sales yards and the cloud.
- Connected the client interface to a serverless NoSQL architecture in Firebase, optimizing query indexing and data transfer.
- Programmed a native Android administrative mobile panel (Kotlin) to automate transactional price updates and multimedia uploads from operational areas.

Professional Web Portfolio | alejandrolopez.dev – Software Engineer

- Developed a high-performance SPA, optimizing load speeds through a scalable architecture and strict separation of concerns (Data Layer vs. UI).
- Designed an interactive JSON data structure viewer for real-time information flow visualization.
- Ensured source code maintainability and quality through automated testing with Vitest and automated pre-commit policies using Husky.

Professional Experience

IT Support Intern, Universidad Estatal a Distancia (UNED) – Puntarenas, Costa Rica Oct 2022 – Dec 2022

- Decreased workstation provisioning time by 40% through infrastructure automated deployments of Windows operating systems using custom-built disk images.
- Secured academic center operational uptime by conducting corrective and preventive hardware and software maintenance on 100% of deployed workstations (+20 devices).
- Mitigated and troubleshooted logical (LAN) and physical connectivity incidents, guaranteeing operational security for staff and students.

Education

Universidad Técnica Nacional (UTN), Bachelor & Licentiate Degree (In progress, 3rd year) in Information Technology Engineering – Costa Rica Jan 2023 – present

- Associate Technical Degree Completed. Core coursework: Advanced Computer Networks, Network Design, Cybersecurity, Database Systems.

Colegio Técnico Profesional de Puntarenas (CTPP), Middle Technical Degree in Computer Networks – Costa Rica Feb 2020 – Dec 2022

Languages

Spanish: Native

English: B2 (Upper-Intermediate / Technical fluency for corporate environments)